

Inspirations into HRH Modelling and Optimization

Reflections

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1 Stock-and-Flow Model (Supply Projection)

1.1 Overview

The Stock-and-Flow Model is a foundational framework in health workforce planning that simulates the evolution of workforce numbers over time. It estimates the net change in health workers by integrating inflows (training outputs, migration, re-entrants) and outflows (retirement, attrition, migration, mortality). This method is widely used by the World Health Organization (WHO), the OECD, and national health ministries for forecasting purposes.

“Workforce planning models typically apply stock-and-flow approaches, accounting for supply flows, losses, and inflow scenarios to estimate future availability.” — WHO, *Global Strategy on Human Resources for Health: Workforce 2030*, 2016

1.2 Mathematical Structure

The general form of the model for cadre k in region r at time t :

$$W_{k,r}(t+1) = W_{k,r}(t) + G_{k,r}(t) + R_{k,r}(t) + I_{k,r}(t) - [\alpha_{k,r}(t) \cdot W_{k,r}(t) + M_{k,r}^{out}(t) + D_{k,r}(t)] + M_{k,r}^{in}(t)$$

Where:

- $W_{k,r}(t)$: workforce stock
- $G_{k,r}(t)$: graduates entering workforce
- $R_{k,r}(t)$: returning professionals
- $I_{k,r}(t)$: foreign-recruited professionals
- $\alpha_{k,r}(t)$: attrition rate
- $M_{k,r}^{out}(t), M_{k,r}^{in}(t)$: migration flows
- $D_{k,r}(t)$: deaths or loss from illness

1.3 Inflow Components

1. Education and Training Outputs

Derived from national accreditation bodies, nursing and medical schools.

Indicator: Graduation rate per cadre and region

Source: WHO National Health Workforce Accounts (NHWA), UNESCO Institute for Statistics

2. Re-entrants

Professionals returning to practice after sabbatical, career changes, or domestic duties.

Often underestimated yet vital during emergencies and health system reform.

3. International Recruitment / Immigration

WHO tracks “health worker mobility” trends between countries.

Net immigration affects national balance sheets in LMICs and HICs differently.

1.4 Outflow Components

1. Attrition

Includes voluntary resignation, career switching, and retirement.

Often modeled as:

$$\alpha_{k,r}(t) = \alpha_{base} \cdot (1 + \beta \cdot \text{workload stress})$$

Sources: MoH HR dashboards, survey-based workload indexes

2. Migration (Outflow)

Frequently driven by better pay, working conditions, or training opportunities abroad.

Data Sources: OECD migration databases, bilateral HRH agreements, WHO Global Code on International Recruitment

3. Mortality & Morbidity

Especially relevant during health crises (e.g., COVID-19, Ebola) when health workers face higher risk.

WHO has documented elevated workforce mortality during pandemics (WHO, 2021).

1.5 Model Uses & Applications

- Workforce projections (short-, medium-, long-term)
- Assess sufficiency to meet UHC targets
- Crisis response simulations (e.g., surge hiring during epidemics)
- Scenario testing: e.g., increasing graduate output vs. reducing attrition

1.6 Examples in Practice

- **South Africa:** Used stock-and-flow models in the Human Resources for Health Strategy 2012/13–2016/17 to model nurse and doctor projections by province

- **OECD:** Applies annual stock-and-flow tracking across 38 member countries for policy benchmarking
- **WHO & HRH2030 Project:** Stock-and-flow used as core methodology for global labor market projections in health

1.7 Toolkits & Resources

- WHO National Health Workforce Accounts (NHWA)
- HRH2030 Health Labor Market Analysis Guide (USAID-funded)
- OECD Health Workforce Planning Methodology

2 Needs-Based Health Workforce Demand Estimation Model

This model incorporates epidemiological, demographic, service delivery, and productivity variables. This model is widely recognized for aligning workforce planning with population health needs, rather than historical utilization or fixed ratios.

2.1 Conceptual Foundation

The needs-based model estimates the number of health workers required to deliver essential services based on the health status of the population, service coverage goals, and productivity norms. It is rooted in the principle that health workforce planning should reflect actual health needs, not just demand or supply trends.

“Needs-based planning makes explicit consideration of population health needs using direct measures of health status and the planned number and type of services to address those needs.” — Asamani et al., *Int. J. Environ. Res. Public Health*, 2021

2.2 Mathematical Formulation

For each cadre k and region r , the required number of health workers is:

$$D_{k,r}(t) = \sum_{s \in S_k} \left(\frac{P_r(t) \cdot p_{s,r}(t) \cdot \lambda_{s,k} \cdot C_{s,r}(t)}{\mu_{s,k,r}(t)} \right) \cdot \left(1 + \delta \cdot \frac{DALY_r(t)}{DALY_{global}(t)} \right) \cdot (1 + \gamma_3 \cdot G_r(t))$$

2.2.1 Variables Explained

Symbol	Description
$P_r(t)$	Population size in region r
$p_{s,r}(t)$	Prevalence of condition s
$\lambda_{s,k}$	Frequency of service per case
$C_{s,r}(t)$	Coverage target (e.g., % of population to be served)
$\mu_{s,k,r}(t)$	Productivity (services per worker per year)
$DALY_r(t)$	Disability-Adjusted Life Years in region r
$G_r(t)$	Gini coefficient (inequality index)
δ, γ_3	Adjustment parameters for disease burden and equity

2.2.2 Dynamic Productivity Adjustment

Productivity is influenced by technology, burnout, and life expectancy:

$$\mu_{s,k,r}(t) = \mu_{s,k}^{\text{base}} \cdot \left(1 + \gamma_1 \cdot \frac{LE_r(t) - LE_{ref}}{LE_{ref}} - \eta \cdot \text{Burnout}_{k,r}(t) \right)$$

Where:

- $LE_r(t)$: Life expectancy in region r
- γ_1 : Tech improvement coefficient
- η : Burnout penalty factor

2.3 Data Sources

Variable	Common Sources
Population & Demographics	National Census, UN DESA
Disease Prevalence	DHS, WHO Global Health Observatory
DALYs	IHME Global Burden of Disease
Coverage Targets	National Health Strategic Plans
Productivity Norms	WHO Service Availability and Readiness Assessments (SARA)
Gini Coefficient	World Bank, UNDP Human Development Reports
Life Expectancy	WHO, World Bank

2.4 Model Applications

- Maternal and newborn care: Estimating midwives and obstetricians needed based on fertility rates and ANC coverage goals
- NCD management: Projecting demand for physicians and nurses based on diabetes, hypertension prevalence
- Sub-national planning: Stratifying demand by rural/urban, age structure, and disease burden

2.5 Advantages

- Aligns workforce planning with UHC and SDG targets
- Sensitive to epidemiological transitions and demographic shifts
- Enables equity-focused planning using Gini and DALY adjustments
- Supports scenario analysis for policy interventions

2.6 Limitations

- Requires high-quality, disaggregated data
- Sensitive to assumptions about productivity and service standards
- May underestimate demand if health status improves rapidly without adjusting prevalence

“Despite its conceptual strength, needs-based planning remains underutilized due to data intensity and lack of open-access tools.” — Lee et al., *Human Resources for Health*, 2024

2.7 References

Asamani, J. A., Christmals, C. D., & Reitsma, G. M. (2021). Advancing the Population Needs-Based Health Workforce Planning Methodology: A Simulation Tool for Country Application. *International Journal of Environmental Research and Public Health*, 18(4), 2113. <https://doi.org/10.3390/ijerph18042113>

Lee, J.T., Crettenden, I., Tran, M. et al. Methods for health workforce projection model: systematic review and recommended good practice reporting guideline. *Hum Resour Health* 22, 25 (2024). <https://doi.org/10.1186/s12960-024-00895-z>

3 Workforce Gap Model

3.1 Conceptual Overview

The Workforce Gap Model is a core analytical tool used to assess whether the current and projected health workforce supply meets the estimated demand. It is typically the third pillar in health workforce projection frameworks—following supply-side and demand-side analyses—and is essential for identifying shortages, surpluses, and distributional imbalances.

“Gap analysis is a critical component of workforce planning, enabling policymakers to quantify mismatches between supply and demand and to prioritize interventions.” — Lee et al., *Human Resources for Health*, 2024

3.2 Mathematical Formulation

For cadre k in region r at time t :

3.2.1 Gap Equation:

$$\Delta_{k,r}(t) = D_{k,r}(t) - W_{k,r}(t)$$

Where:

- $D_{k,r}(t)$: Demand (needs-based estimate)
- $W_{k,r}(t)$: Workforce supply (stock-and-flow projection)
- $\Delta_{k,r}(t)$: Net gap (positive = shortage; negative = surplus)

3.2.2 Coverage Ratio:

$$\text{Coverage}_{k,r}(t) = \frac{W_{k,r}(t)}{D_{k,r}(t)}$$

3.2.3 Shortage Flag:

$$\text{Shortage}_{k,r}(t) = \begin{cases} 1 & \text{if } \frac{\Delta_{k,r}(t)}{D_{k,r}(t)} > \theta_k \\ 0 & \text{otherwise} \end{cases}$$

Where θ_k is a cadre-specific threshold (e.g., 10% for physicians).

3.3 Types of Gaps Identified

- **Numerical Shortage:** Total number of workers below required level

- **Skill Mix Imbalance:** Wrong proportion of cadres (e.g., too few nurses vs. doctors)
- **Geographic Maldistribution:** Urban vs. rural disparities
- **Sectoral Imbalance:** Public vs. private sector concentration
- **Temporal Gap:** Lag between training and deployment

“Health workforce challenges are not only about numbers but also about distribution, skill mix, and timing.” — WHO, *Global Strategy on Human Resources for Health: Workforce 2030*, 2016

3.4 Data Requirements

Variable	Source
Workforce stock	HRIS, professional registries
Demand estimates	Needs-based model, DALYs, coverage targets
Population data	Census, UN DESA
Service utilization	DHS, facility records
WHO benchmarks	WHO Global Health Workforce Statistics

3.5 Use Cases

- National HRH strategies: Identifying cadres and regions with critical shortages
- Training policy: Adjusting intake based on projected gaps
- Budgeting: Prioritizing investments in high-gap areas
- Scenario analysis: Testing policy interventions (e.g., task shifting, retention incentives)

3.6 Limitations

- Sensitive to assumptions in supply and demand models
- May not capture qualitative gaps (e.g., competency, motivation)
- Requires high-quality, disaggregated data

“Workforce projection models often rely on imperfect data with limited granularity at the local level.” — Lee et al., *Human Resources for Health*, 2024

4 Optimization Model

4.1 Conceptual Overview

The Optimization Model is designed to determine the most efficient allocation of health workers across cadres, regions, and services—within real-world constraints such as budgets, training capacity, and skill mix. Unlike forecasting models, optimization focuses on decision-making: how to best meet health service goals given limited resources.

“Workforce optimization seeks to achieve the best health workforce management outcomes within a given fiscal context.” — Alawode et al., *Human Resources for Health*, 2025

4.2 Mathematical Formulation

Objective Function: Minimize cost and unmet demand:

$$\min \sum_{k,r} [c_k \cdot W_{k,r}(t) + \phi_k \cdot \Delta_{k,r}(t)^2]$$

Subject to Constraints:

1. Budget Constraint:

$$\sum_{k,r} c_k \cdot W_{k,r}(t) \leq B_r(t)$$

2. Skill Mix Constraint:

$$\frac{W_{\text{nurse},r}(t)}{W_{\text{physician},r}(t)} \geq \rho_{\min}$$

3. Training Capacity:

$$G_{k,r}(t) \leq \Gamma_{k,r}(t)$$

Where:

- c_k : cost per cadre
- ϕ_k : penalty for unmet demand
- $B_r(t)$: budget in region r
- $\rho_{\min}$: minimum nurse-to-physician ratio
- $\Gamma_{k,r}(t)$: maximum training output

4.3 Key Inputs

- Workforce cost data (salaries, training, incentives)

- Demand estimates from needs-based models
- Supply projections from stock-and-flow models
- Policy targets (e.g., UHC coverage, skill mix ratios)
- Training capacity by cadre and region

4.4 Data Sources

Variable	Source
Cadre costs	Ministry of Health, Finance, WHO Global Health Expenditure
Skill mix ratios	WHO benchmarks, national HRH strategies
Budget ceilings	National health accounts, IMF reports
Training capacity	Accreditation bodies, education ministries

4.5 Applications

- National HRH planning: Allocate workforce across provinces based on cost-effectiveness
- Scenario testing: Compare outcomes under different budget or training expansion scenarios
- Task shifting: Evaluate substitution of lower-cost cadres (e.g., nurses for physicians)
- Emergency response: Rapid reallocation during epidemics or humanitarian crises

“Optimization models are essential for aligning workforce deployment with health system goals, especially in resource-constrained settings.” — WHO, *Models and Tools for Health Workforce Planning and Projections*, 2010

4.6 Limitations

- Requires high-quality cost and productivity data
- May oversimplify complex human factors (e.g., motivation, retention)
- Sensitive to assumptions about service equivalence across cadres

“Optimization should be complemented by qualitative insights and stakeholder engagement to ensure feasibility and acceptability.” — Sonderegger et al., *Human Resources for Health*, 2021

4.7 Reference

Alawode, G.B., Ajibola, A.R.A., Sanusi, M.S. et al. Optimizing the health workforce for Universal Health Coverage: a framework for analysis and action. *Hum Resour Health* 23, 27 (2025). <https://doi.org/10.1186/s12960-025-01000-8>